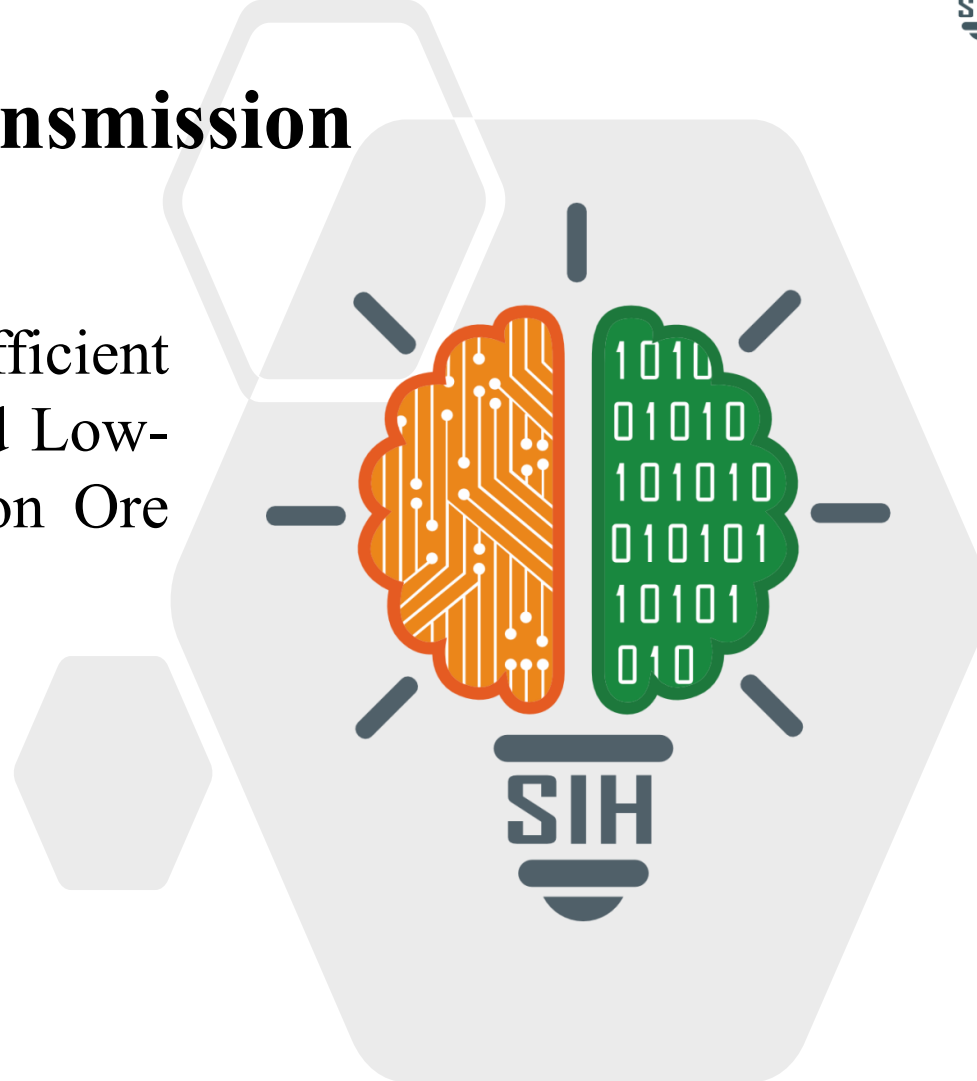


SMART INDIA HACKATHON 2026



Lost in transmission

- **Problem Statement ID – 26007**
- **Problem Statement Title-** Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.
- **Theme- Smart Automation**
- **PS Category- Hardware**
- **Team ID- 176278**
- **Team Name- Lost in transmission**



Problem in hand

Regulatory Halts: DGMS mandates force fleet shutdowns during fog, delaying millions of tons of ore annually.

Sensor Failure: Standard 3D LiDAR fails in dense fog due to the scattering of visible light.

Fatal Terrain Risks: Low visibility turns unpaved roads with 8-10% gradients and 50-100m cliffs into invisible hazards.

Operational Paralysis: Monsoon fog reduces visibility to 3-5 meters, halting daily operations for 4+ hours.

Why we stand out?

Fog-Independent Obstacle Confirmation

Cross-validates radar and thermal detections to improve reliability in dense fog.

Solution “Minesight” is a 3D mmWave radar + thermal fusion system with V2V cooperative perception for obstacle, cliff, edge, and worker detection in monsoon fog.

- Detect 3D obstacles (x, y, z coordinates) using RD-60 60GHz mmWave point cloud
- Detect approaching trucks and obstacles in fog using distributed radar and thermal sensor fusion
- Display real-time 3D point cloud, distance, velocity, and confidence on driver dashboard
- Enable vehicle-to-vehicle (V2V) position sharing via LoRa mesh network for situational awareness

Cooperative V2V Hazard Sharing

Propagates detected hazards to approaching vehicles.

Predictive TTC Risk

Predicts collision risk from distance + relative velocity.

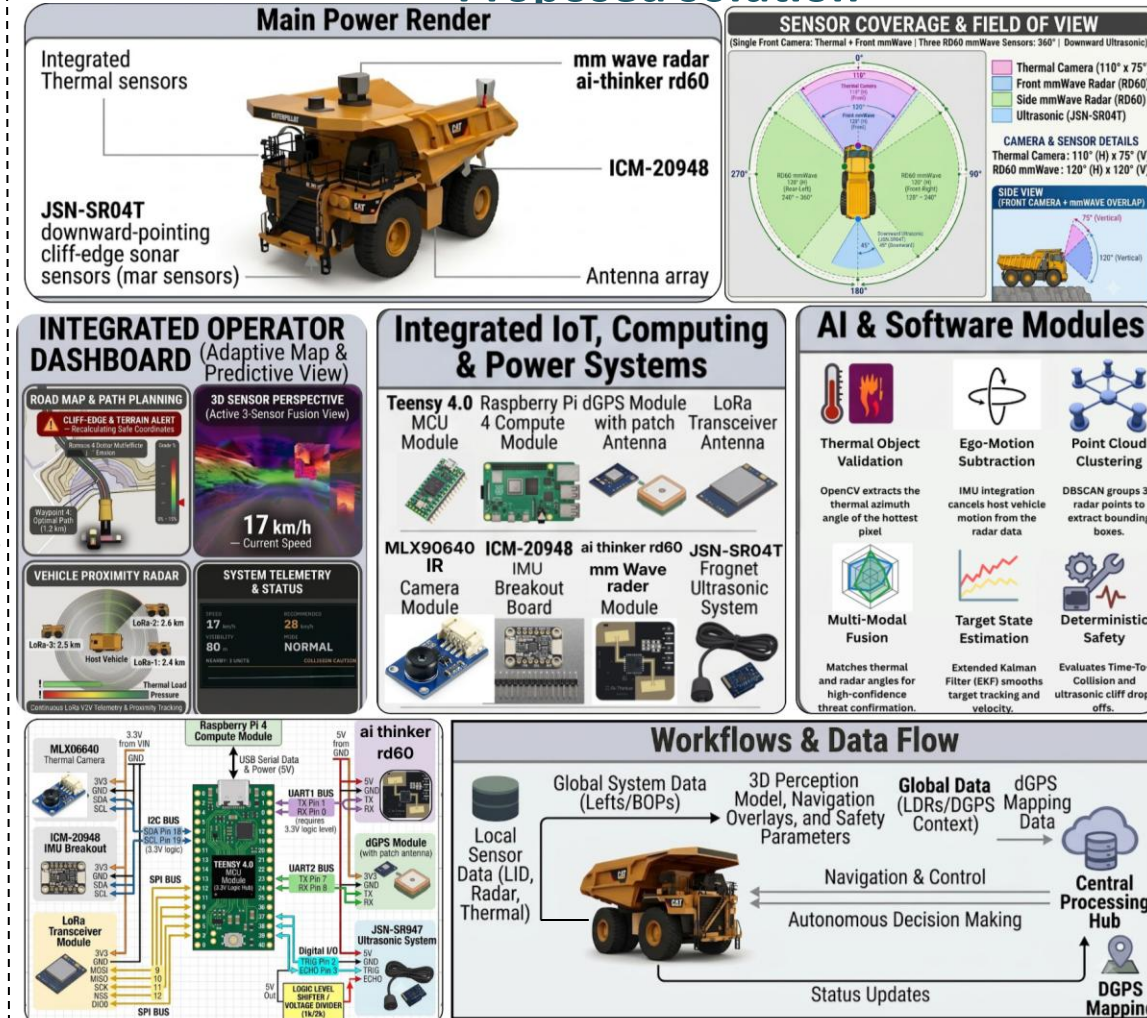
Risk-Adaptive Speed Advisory

Recommends safe operating speed instead of only warning.

Dynamic Local 3D Scene

Tracks nearby objects and motion for predictive decisions.

Proposed solution



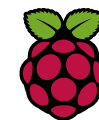
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TECHNICAL APPROACH

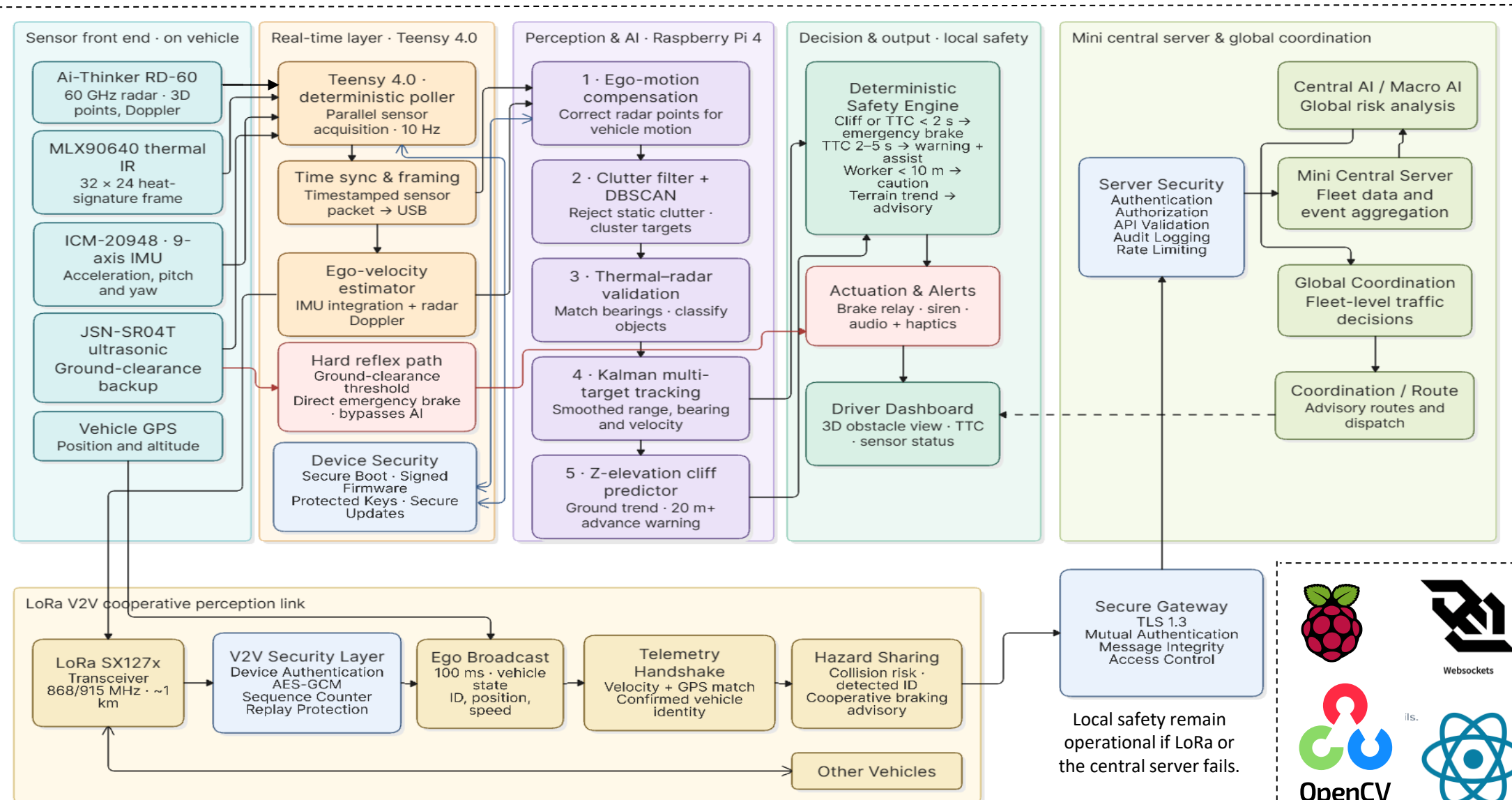
Technical stack



JavaScript



arm



Feasibility

Technical Feasibility



C/C++ edge architecture on accessible hardware supports low-latency processing without relying on heavy ML models.

Operational Feasibility



mm Wave radar and decentralized LoRa mesh maintain local collision avoidance in low visibility, independent of central mine networks.

Economic Feasibility



Reduces fog-related vehicle stoppages, **minimizing production losses** and haul-cycle delays while **integrating with existing HEMMs** without major modifications.

Regulatory Feasibility



TTC-based braking logic and ultrasonic cliff detection align with DGMS safety guidelines for heavy earth-moving machinery

Challenges



Multi-Sensor Synchronization

- Radar, thermal, IMU, GNSS, and odometry operating at different rates



Reliable Detection in Extreme Fog

- Maintaining reliable detection under severe fog, rain, and dust



Accurate 3D Position Estimation

- Converting radar/thermal measurements to 3D positions on steep haul roads



False Detection & Sensor Conflicts

- Resolving false targets and associating object data across sensors



Real-World Vehicle Integration

- Mounting, waterproofing, power supply, and HEMM system integration

Solution



Sensor Synchronization

- Timestamped multi-sensor fusion for consistent real-time data.



Extreme-Fog Perception

- Radar + thermal redundancy maintains detection under degraded visibility.



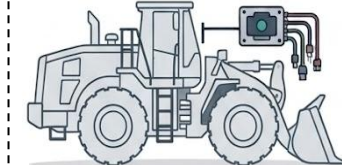
3D Localization

- IMU-assisted coordinate correction for moving and inclined vehicles.



False Detection Handling

- Object association and confidence scoring reduce conflicting detections.



HEMM Integration

- Modular edge architecture enables gradual transition from prototype to vehicle deployment.

Economic Benefits

Reduced Operational Losses Predictive hazard detection and safer vehicle coordination can reduce collision-related disruption, improve vehicle utilization, and support more consistent ore haulage during low-visibility conditions.

Safety Benefits

Improved Collision Awareness. Multi-sensor detection, TTC-based risk assessment and V2V hazard alerts provide earlier warning of nearby vehicles, workers and obstacles in low visibility.

Operational Benefits

Improved Haulage Continuity. Risk-aware speed guidance can help maintain safe vehicle movement during fog, reducing unnecessary stoppage and operational disruption.

1 DUMPER / VEHICLE OPERATOR



- Reduced Driver Workload
- Earlier Hazard Awareness
- Safer Fog Operation

2 NEARBY VEHICLES



- Earlier Warning
- Reduced Blind-Spot Risk
- Fleet-Wide Situational Awareness

3 MINE CONTROL / SUPERVISOR



- Faster Decisions
- Better Fleet Coordination
- Improved Incident Awareness

4 MINE MANAGEMENT

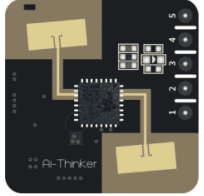


- Reduced Downtime
- Improved Haul-Cycle Continuity
- Better Fleet Utilization

Our promise

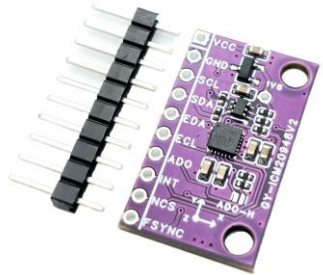
MineSight aims to keep mine vehicles moving safely when visibility falls - reducing unnecessary stoppage while improving collision awareness and fleet coordination

Component datasheets



Ai-Thinker Rd-60 (60 GHz mmWave Radar): 60 GHz FMCW radar, 120° H/V FOV, point-cloud output. Reference: [Ai-Thinker Rd-60 documentation](#)

Melexis MLX90640 (Thermal Array): Component for capturing dynamic heat signatures. Reference: <https://www.melexis.com/en/product/mlx90640/>



TDK InvenSense ICM-20948 (9-Axis IMU): 9-axis motion tracking sensor with 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer. Reference: [TDK ICM-20948](#)

JSN-SR04T (Ultrasonic): Waterproof industrial sensor for close-range edge monitoring. Reference: <https://www.electroschematics.com/jsn-sr04t/>



Thermal Camera & Low-Visibility Object Detection

Academic Paper on Thermal Robustness: "*Image Translation from Thermal to RGB for Vehicle Detection in Low-Light Conditions*". Highlights that while standard optical detection fails in low-visibility or glaring conditions, infrared imaging remains highly robust by tracking thermal radiation (engine heat). Reference: <https://arxiv.org/pdf/2209.09808>

3D Point Cloud Mapping using mmWave Radar

Enhancing Radar Point Clouds in Adverse Weather: Validates supervised learning (mmEMP) for 60GHz FMCW radar, proving it is a robust, low-cost alternative to high-performance LiDAR in dense fog and smoke. Reference: <https://arxiv.org/html/2404.17229v1>

Deep Learning for mmWave Point Clouds: Research on *Radar to Point Cloud (R2P)* models to convert sparse radar inputs into dense 3D vehicle shapes. Reference: https://www.researchgate.net/publication/362230459_R2P_A_Deep_Learning_Model_from_mWave_Radar_to_Point_Cloud

Curated Algorithms: Awesome mmWave Radar Perception: <https://github.com/Armorhtk/awesome-mmwave-radar-perception> | Awesome 3D Detection with 4D Radar: <https://github.com/liuzengyun/Awesome-3D-Detection-with-4D-Radar>

LoRa Mesh & V2V Cooperative Perception

Network Collision Physics (Semtech LoRa Architecture): Statutory guidelines for scaling a 10Hz update rate across an entire fleet while actively managing radio channel collision limits. Reference: <https://www.semtech.com/uploads/technology/LoRa/lora-and-lorawan.pdf>

V2V Mesh Logic (painlessMesh): Standard GitHub library for establishing self-organizing, decentralized ad-hoc networks on microcontrollers without a central router. Reference: <https://github.com/gmag11/painlessMesh>